

Document reference		NPS/ 001 / 019					
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NPS/001/019 – Technical Specification for Low Voltage Conductor Shrouding Materials

1. Purpose

This document details the requirements for temporary conductor shrouding for use on low voltage overhead lines distribution network of CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/001/019	1.0	Technical Specification for Low Voltage Conductor Shrouding Materials

2. Scope

This document specifies temporary insulating material with the objective of providing protection from inadvertent contact with exposed live conductors in situations where safe-working clearances cannot be achieved. It must be suitable for use during both proximity working of company personnel and third party work situations.

3. Technical Requirements

3.1 General Requirements

The requirement is for low voltage overhead line conductor shrouding in accordance with Class 2 as detailed in ENA TS 43-103 Issue 1. This must be suitable for use by company personnel during live low voltage work and provide proximity protection for third party proximity work situations.

Shrouding shall be in the form of split tubing and be designed to remain attached to the conductors in adverse weather conditions. The mechanical and electrical properties shall be retained through the temperature range of -20°C and $+70^{\circ}\text{C}$. It shall be manufactured in such a manner as to be free from cracks, holes, foreign bodies or other physical defects liable to adversely affect its performance under service conditions.

The shrouding shall be suitable for use where the operating voltage of the line does not exceed 500V rms between line conductors or 250V rms between any conductor and earth.

Split tubing shall have a minimum internal diameter of 16mm, supplied in manageable lengths (preferred length 3 meters) with a means of joining multiple lengths together (one for each length). It shall be marked with the following information:

- a) Maximum safe working voltage (500V)
- b) Year and month of manufacture

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c) Manufacturer's name or mark

d) Class 2

3.2 Type Test Requirements

Type test shall be in accordance with ENATS 43-103 Issue 1. Appendix 1 contains a self certification conformance declaration and shall be completed by the supplier.

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4. References

4.1 External Documentation

Reference	Title
ENA TS 43-103: Issue 1 - 1995	Overhead Line Temporary Shrouding Materials

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards and Assessments Engineer	S Salkeld	13/09/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Purchasing Manager	G Bolden	20/09/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager	G Hammel	17/09/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	12/10/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager	C Holdsworth	17/09/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	17/09/07

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Appendix 1 - SELF CERTIFICATION CONFORMANCE DECLARATION

Low voltage overhead line temporary shrouding materials shall comply with the latest issues of the relevant international and international standards.

ENATS 43-103 Issue 1: 1995 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses of the aforementioned standard relevant to Low voltage overhead line temporary shrouding materials for use on the CE Electric UK distribution network.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each item or variant submitted.

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Appendix 1

Low voltage overhead line temporary shrouding materials

• ENATS 43-103 Issue 1: 1995			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
4.4	Marking		
5.1	General Requirements		
5.2	Mechanical tests		
5.3	Electrical tests		
5.4	Flammability test		